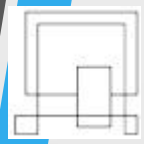
VIA University College



Programming 3

Autumn 2026

Spring Boot

- SpringBoot = a Sub-Project of SpringFramework

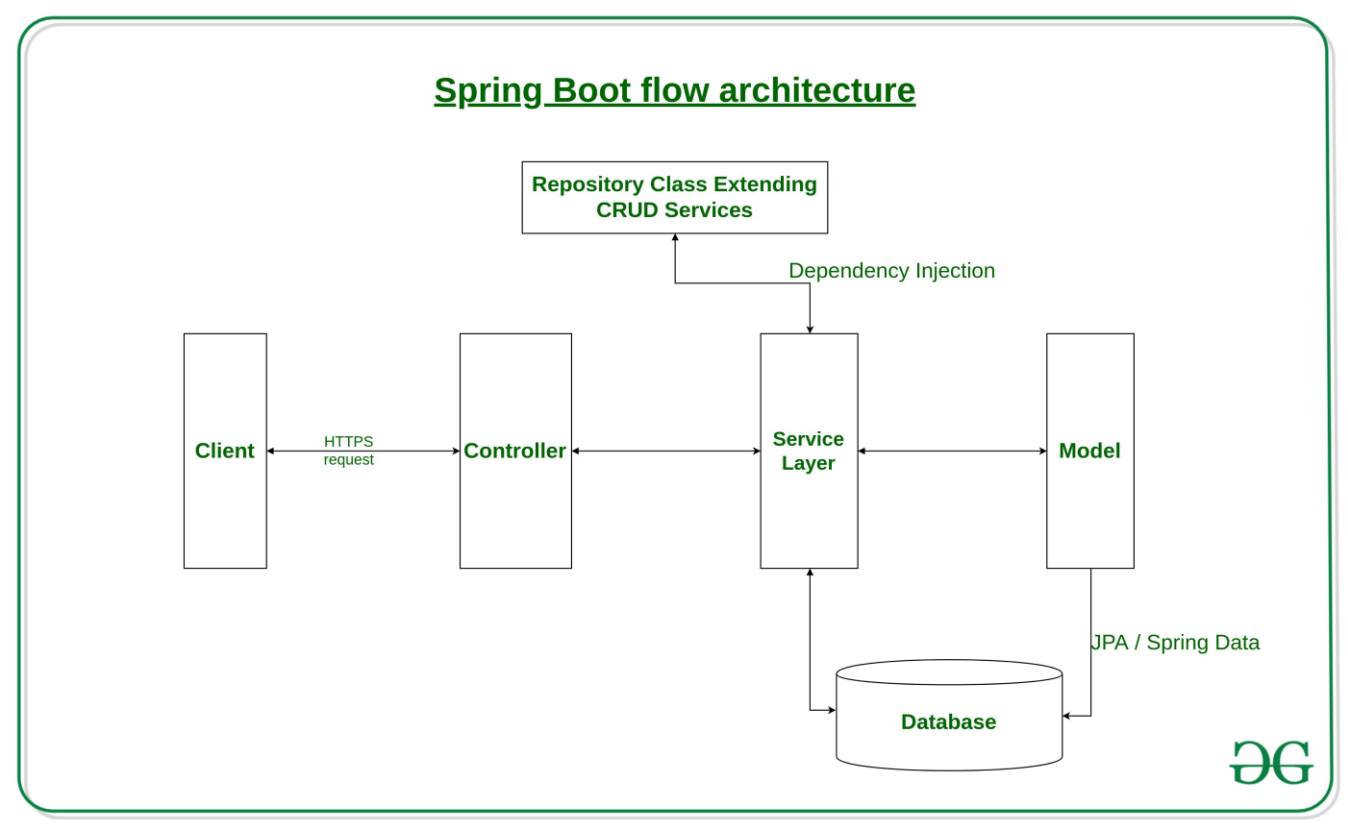
- From Wikipedia:
  - The Spring Framework is an **application framework** and **inversion of control container** for the Java platform.
  - can be used by any Java application ..
  - .. extensions for web applications.
  - … popular in the Java community .. alternative EJB.
  - open source

Spring Boot

- Starter POMS
  - Automatic configuration with sensible defaults
  - Example: spring-boot-starter-web
- Features
  - Monitoring
  - Metrics
-



Spring Boot flow architecture



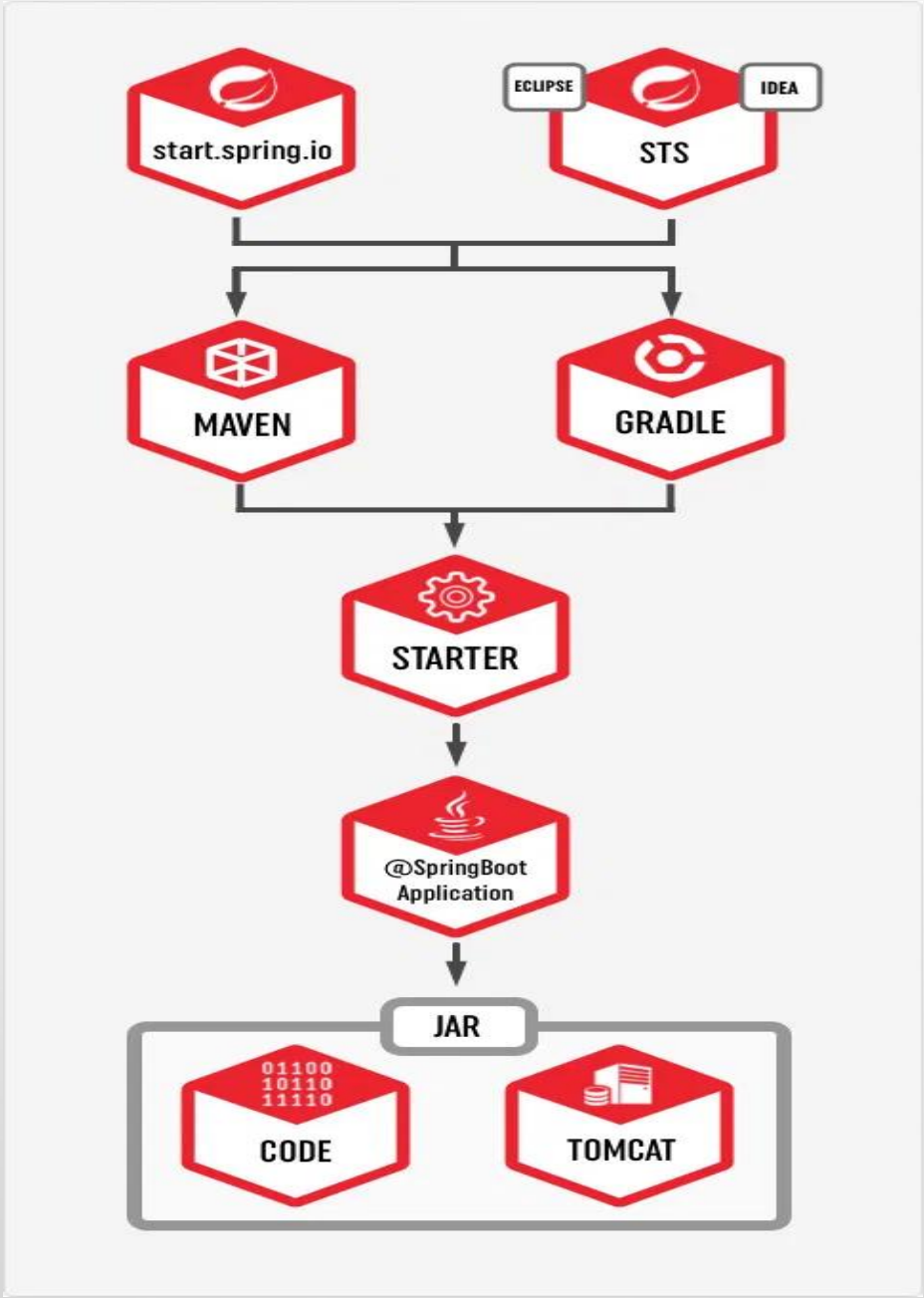
https://www.geeksforgeeks.org/introduction-to-spring-boot/

Spring Boot Goals

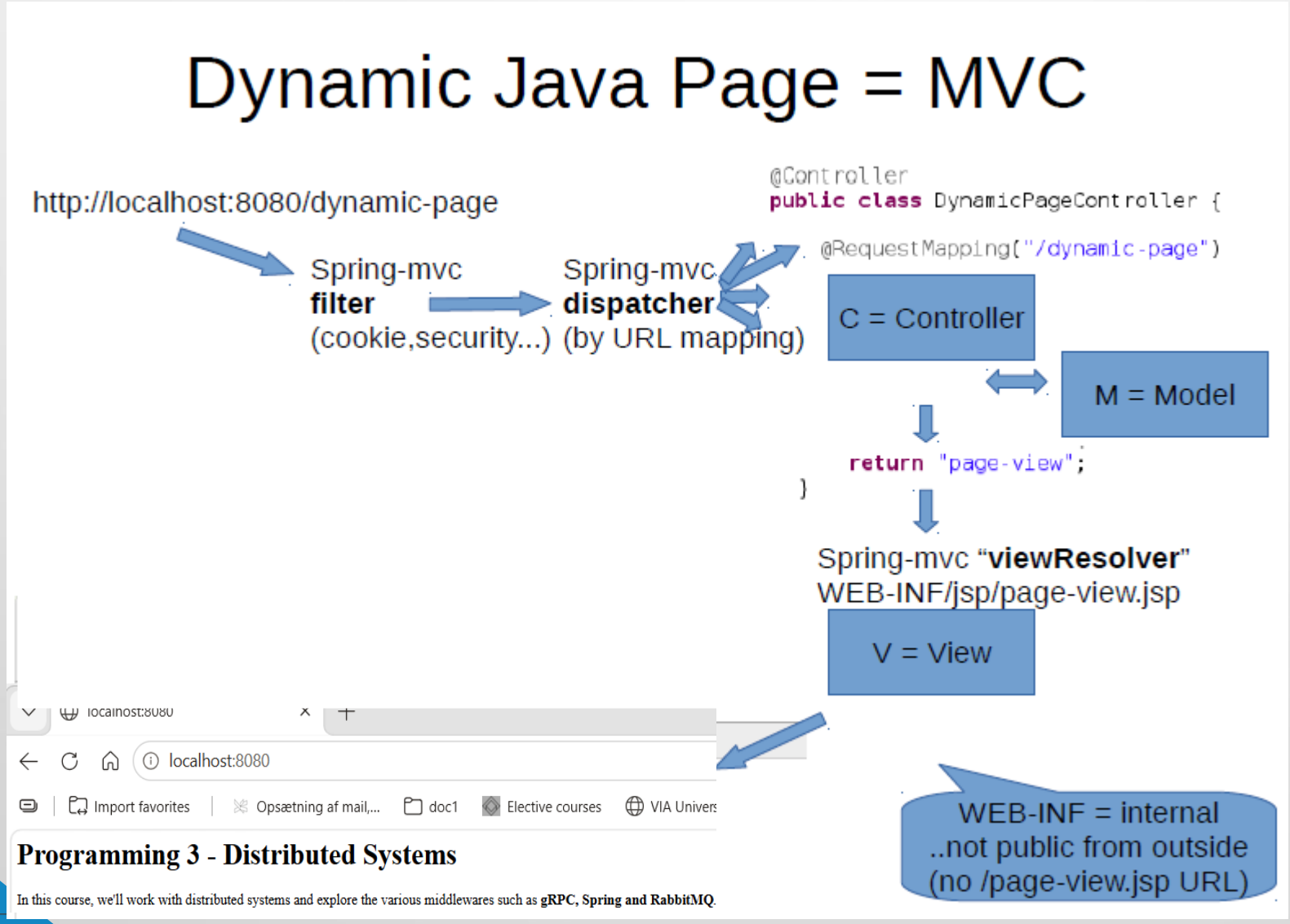
- To avoid complex XML configuration in Spring
- To develop a production ready Spring applications in an easier way
- To reduce the development time and run the application independently
- Offer an easier way of getting started with the application

How does it work?

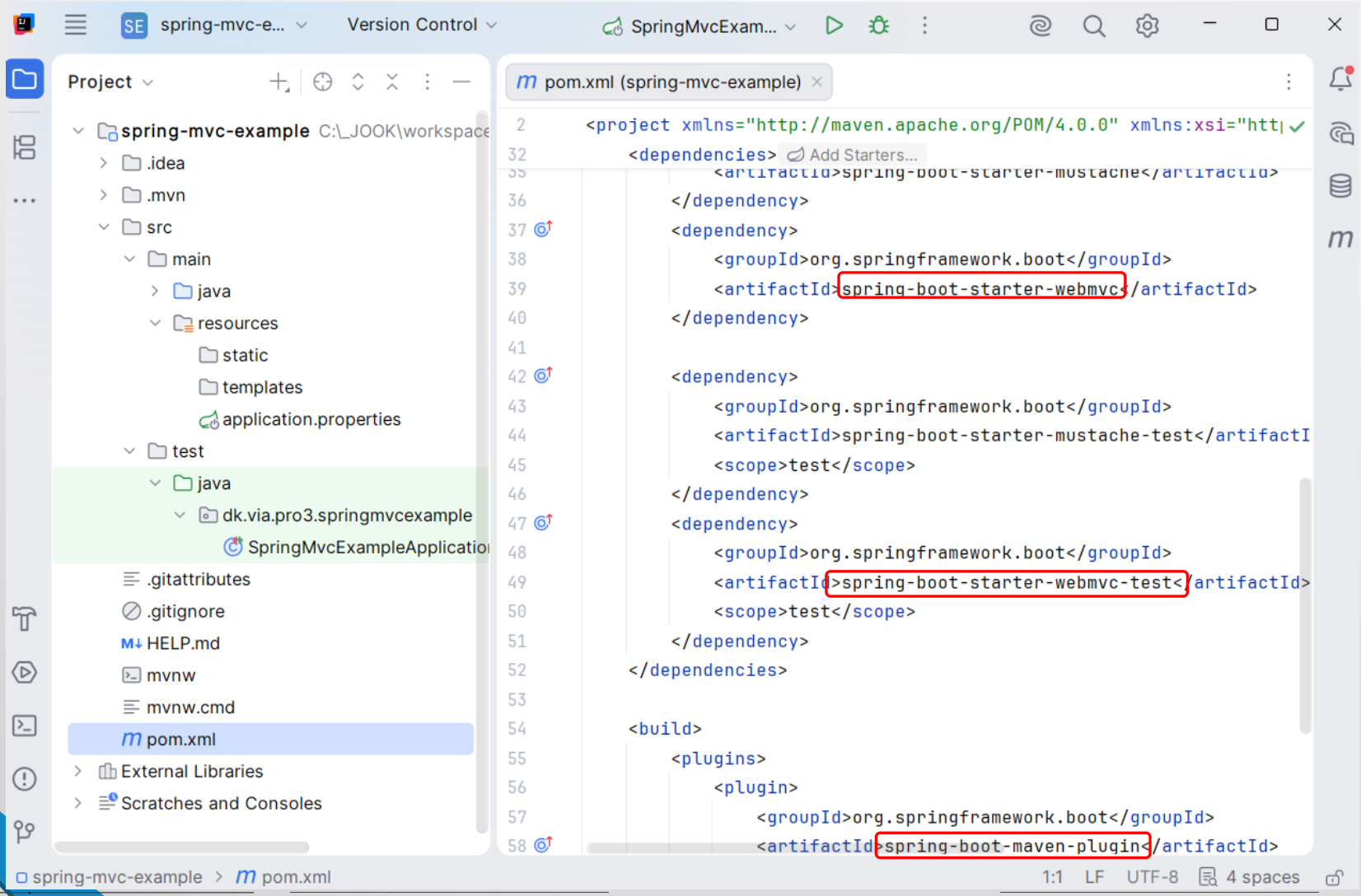
- Spring Boot automatically configures your application based on the dependencies you have added to the project using **@EnableAutoConfiguration** annotation
- The entry point of the spring boot application is the class contains **@SpringBootApplication** annotation and the main method.
- Spring Boot automatically scans all the components included in the project through using **@ComponentScan** annotation


start.spring.io
ECLIPSE
IDEA
STS
MAVEN
GRADLE
STARTER
@SpringBoot
Application
JAR
01100
10110
11110
CODE
TOMCAT

Spring MVC


Dynamic Java Page = MVC
http://localhost:8080/dynamic-page
Spring-mvc
filter
(cookie,security...)
Spring-mvc
dispatcher
(by URL mapping)
@Controller
public class DynamicPageController {
@RequestMapping("/dynamic-page")
C = Controller
M = Model
return "page-view";
}
Spring-mvc "viewResolver"
WEB-INF/jsp/page-view.jsp
V = View
localhost:8080
Import favorites
Opsætning af mail,...
doc1
Elective courses
VIA Univers
Programming 3 - Distributed Systems
In this course, we'll work with distributed systems and explore the various middlewares such as gRPC, Spring and RabbitMQ.
WEB-INF = internal
..not public from outside
(no /page-view.jsp URL)

Spring Boot
Source Code Structure



With REST API (Session 5, 6)

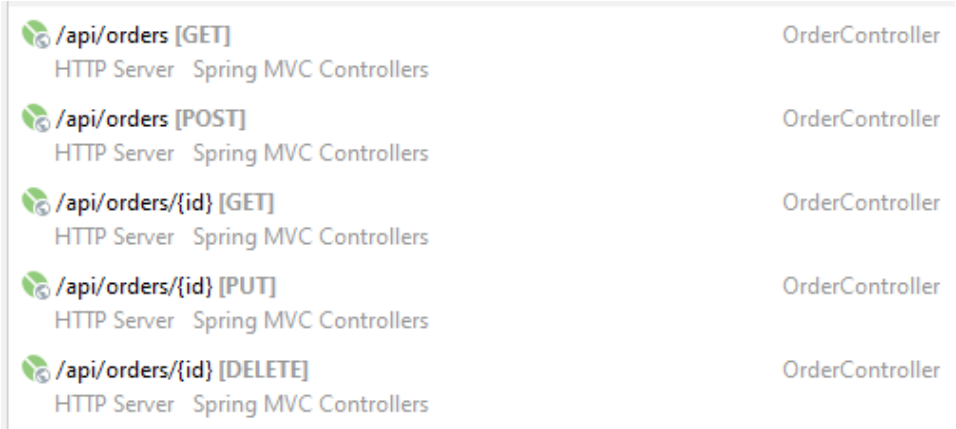
Add REST/Json Web API



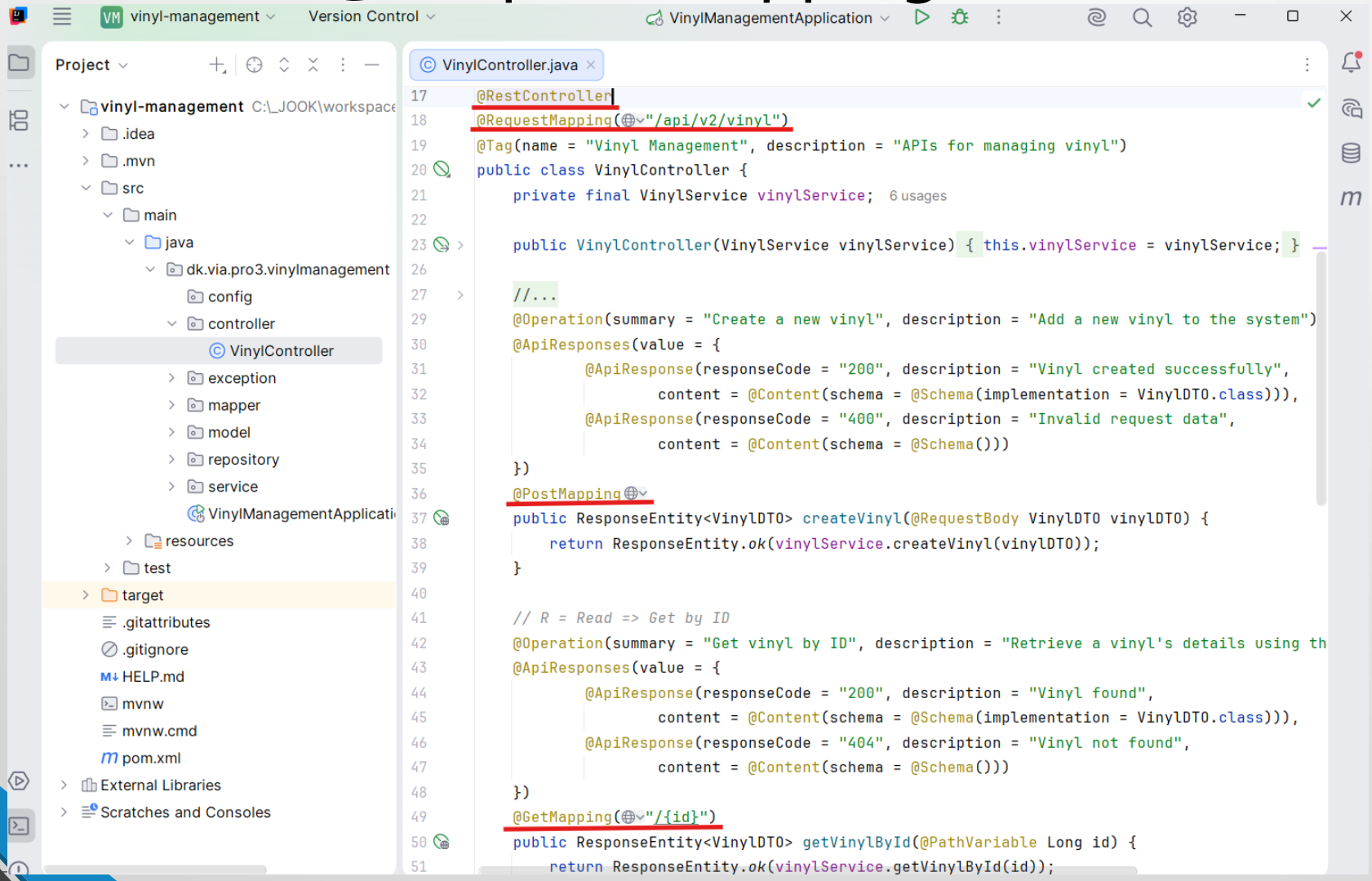
Using "Built-in" spring-web-mvc	Bind to JAX-RS standard
specific annotations	standard annotations
@RequestMapping	@Path @GET
@RequestBody ...	@ ...

HTTP Protocol – RESTful actions

- @GetMapping corresponds to HTTP GET
- @PostMapping corresponds to HTTP POST
- @PutMapping corresponds to HTTP PUT
- @PatchMapping corresponds to HTTP PATCH*
- @DeleteMapping corresponds to HTTP DELETE



Using @RestController @RequestMapping



Live Coding/Hands-on/Exercises